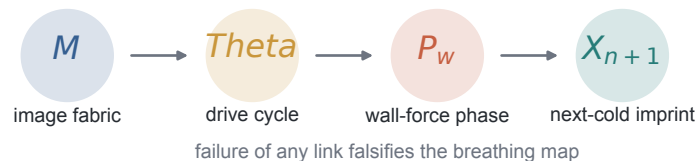
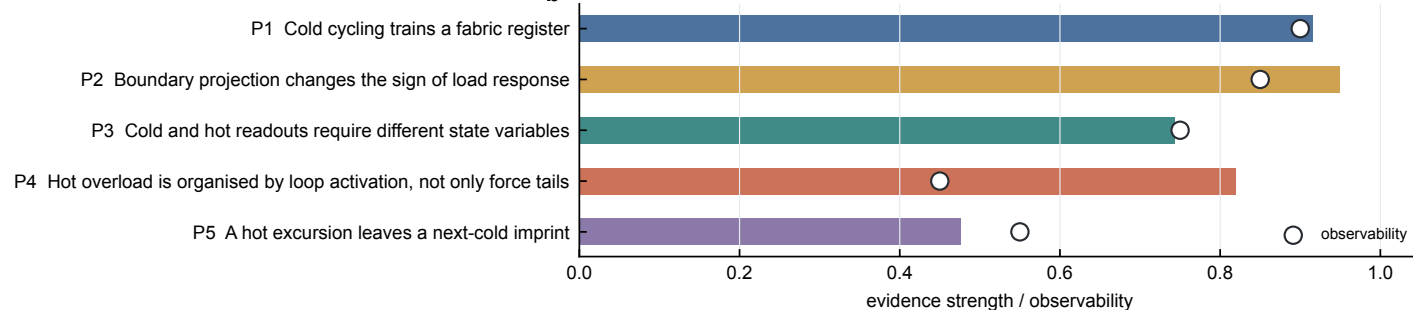


a falsifiable experimental ladder

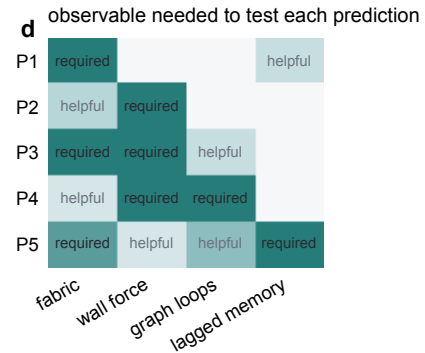


b what the present DEM already constrains



e decisive failure modes

- P1** No persistent rise in fabric/contact-survival proxy under repeated identical cycles
- P2** All boundary protocols collapse onto one monotonic pressure-growth curve
- P3** Residual cold load and transient hot overload collapse onto the same scalar density or pressure coordinate
- P4** Force-tail metrics predict overload after graph-loop embedding is destroyed or randomised
- P5** Hot-phase excursions do not predict any next-cold fabric or loop-memory observable



c hardest test is loop-resolved force topology

